

TAMIL NADU STATE BOARD - STANDARD 9 MATHEMATICS

CHAPTER 6: TRIGONOMETRY

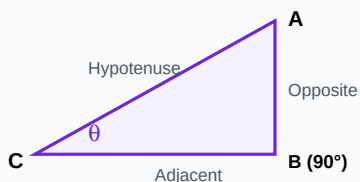
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Mathematics
Chapter Name:	Trigonometry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

UNIT ROADMAP: INTRODUCTION TO TRIGONOMETRIC RATIOS



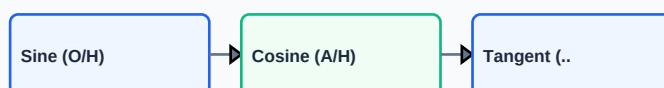
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Right-Angled Triangle	A triangle containing a 90-degree angle, forming the basis of trigonometric definitions.
Trigonometric Ratios	Ratios of the sides of a right-angled triangle relative to an acute angle θ .
Complementary Angles	Two angles whose sum is exactly 90 degrees. Trigonometric ratios show key symmetric relationships here.
Reciprocal Ratios	The inverse trigonometric ratios: cosec θ , sec θ , and cot θ .

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

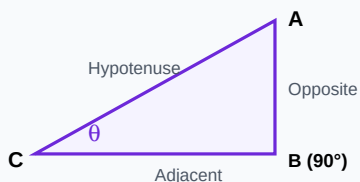
GLOSSARY: PRIMARY TRIGONOMETRIC RATIOS



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: RIGHT TRIANGLE OPPOSITE/ADJACENT RATIOS



Essential Formulas, Laws & Identities:

- $\sin \theta = \text{Opposite} / \text{Hypotenuse}$
- $\cos \theta = \text{Adjacent} / \text{Hypotenuse}$
- $\tan \theta = \text{Opposite} / \text{Adjacent} = \sin \theta / \cos \theta$
- Complementary rules: $\sin(90 - \theta) = \cos \theta$ | $\cos(90 - \theta) = \sin \theta$
- Reciprocal relationships: $\operatorname{cosec} \theta = 1/\sin \theta$ | $\sec \theta = 1/\cos \theta$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: In a right triangle ABC, opposite side = 3 cm, adjacent side = 4 cm. Find $\sin \theta$ and $\cos \theta$.

Step-by-step Solution:

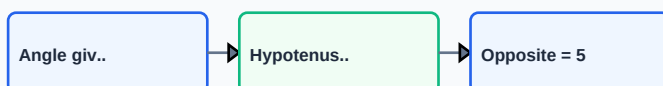
Find hypotenuse using Pythagoras: $\text{hyp} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5 \text{ cm}$. $\sin \theta = 3/5 = 0.6$ $\cos \theta = 4/5 = 0.8$.

Example 2: Evaluate $\sin 30$ degrees + $\cos 60$ degrees.

Step-by-step Solution:

Recall values: $\sin 30 = 1/2$, $\cos 60 = 1/2$. Sum = $1/2 + 1/2 = 1$.

WORKED: SOLVING RIGHT TRIANGLE SIDE



HOTS (Higher Order Thinking Skills) Challenge

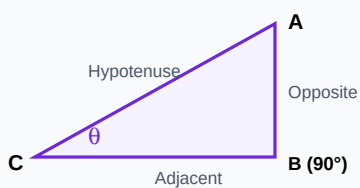
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: If $\tan \theta = 15/8$, find all other trigonometric ratios of θ .

Analytical Solution Strategy:

Given opposite = 15, adjacent = 8. $\text{hyp} = \sqrt{(15^2 + 8^2)} = \sqrt{(225 + 64)} = \sqrt{289} = 17$. $\sin \theta = 15/17$, $\cos \theta = 8/17$.
 $\text{cosec } \theta = 17/15$, $\sec \theta = 17/8$, $\cot \theta = 8/15$.

HOTS: HEIGHT OF TOWER USING ANGLE OF ELEVATION



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find $\cot \theta$ if $\tan \theta = 4/3$.

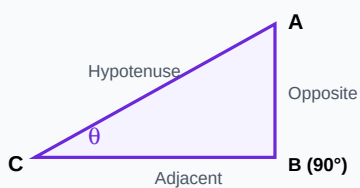
Q2 (2-Mark): State the values of $\sin 45$, $\cos 45$, and $\tan 45$.

Q3 (2-Mark): Verify if $\cos(90 - 30) = \sin 30$.

Q4 (5-Mark): Simplify: $(\sin^2 \theta + \cos^2 \theta)$.

Q5 (5-Mark): Find $\sec \theta$ if $\cos \theta = 1/\sqrt{2}$.

PRACTICE: ANGLE OF ELEVATION RATIOS



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Oceanography

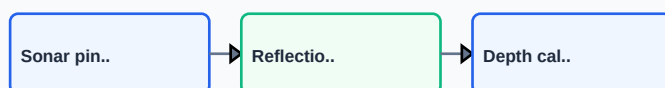
Trigonometric equations help calculate wave heights, high tide peaks, and sea current speeds based on astronomical alignments.

Application Case: Architecture & Surveying

Land surveyors use trigonometry (triangulation) to measure mountain heights and build vertical truss structures.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: TRIGONOMETRY IN SONAR MAPPING



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: TRIGONOMETRY SUMMARY FLOW

